

SI Session 3

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1 Mixture Rate Problem

1.1 Problem 1

A 600-liter tank initially contains 79 liters of a fluid in which 223 grams of a certain chemical have been dissolved. Starting at $t = 0$, fluid containing 30 grams per liter of the same dissolved chemical flows into the tank at 6 liters/min. If the mixture is kept uniform by constant stirring and if the well-stirred mixture flows out of the tank at 3 liters/min, (a) how much of the chemical is in the tank at the end of 10 minutes, and (b) how much of the chemical is in the tank when the tank overflows?

2 Vertical Motion Problem

2.1 Problem 2

At $t = 0$, a 16 pound object is pushed beneath the surface of a calm lake, and then released from rest. Since the object has a buoyancy force of 32 pounds, the object begins to rise after it is released. The resistance (in pounds) of the water is $v/4$, where v is the velocity (in feet/sec) of the rising object. If the object reaches the surface in 8 seconds, what is the velocity of the object when it reaches the surface of the lake?